OCR GCSE Computer Science — **Networks: Types, Topologies and Hardware (J277/01 Section 1.3.1—1.3.2)**. This topic covers the fundamentals of computer networks including LANs and WANs, client-server and peer-to-peer models, star and mesh topologies, network hardware components, and wired versus wireless connections. Exam questions typically ask you to **describe** the difference between LANs and WANs, **compare** client-server and peer-to-peer networks, **explain** the purpose of network hardware, **draw and label** network topologies, and **discuss** advantages and disadvantages of wired versus wireless connections. The 2024 exam included questions about network hardware functions and topology characteristics.

What you must know

This checklist covers the key exam-visible concepts. You should be able to explain each point and apply it to real-world scenarios.

- **Describe** the characteristics of LANs and WANs and give examples of each.
- **Compare** client-server and peer-to-peer network models, including advantages and disadvantages.
- **Explain** the purpose of network hardware: NIC, switch, router, WAP, and transmission media.

- **Draw and describe** star and mesh network topologies.
- **Explain** factors that affect network performance (bandwidth, number of users, interference, etc.).
- **Compare** wired and wireless connections, including advantages and disadvantages.
- **Describe** different connection methods: Ethernet, WiFi, and Bluetooth.
- **Explain** encryption and its role in securing wireless networks.

> **Specification note.** OCR J277 Section 1.3.1 covers network types, topologies, and hardware. Section 1.3.2 covers wired and wireless networks. Questions often present scenarios (school network, home network, business) and ask you to recommend appropriate network configurations with justification.

Toolkit

LAN vs WAN comparison

Feature	LAN (Local Area Network)	WAN (Wide Area Network)
Geographic area	Small area (single building, site, campus)	Large area (cities, countries, worldwide)
Ownership	Usually owned by single organisation	Often uses third-party infrastructure (ISPs, telecoms)
Speed	Generally faster (Gbps typical)	Generally slower (depends on connection)
Cost	Lower setup and maintenance costs	Higher costs (leased lines, equipment)
Examples	School network, office network, home network	The Internet, bank networks spanning cities

Client-server vs peer-to-peer

Feature	Client-server	Peer-to-peer (P2P)
Structure	Central server(s) provide services to clients	All computers are equal; share resources directly
Security	Centralised security, easier to manage	Each device responsible for own security
Backups	Centralised backups on server	Each user backs up own data
Cost	Higher (server hardware, admin staff)	Lower (no dedicated server needed)

Feature	Client-server	Peer-to-peer (P2P)
Scalability	Easy to add more clients	Performance degrades as network grows
Best for	Schools, businesses, organisations	Small home networks, file sharing

Network hardware summary

Hardware	Purpose
NIC (Network Interface Card)	Allows a device to connect to a network; has a unique MAC address
Switch	Connects devices within a LAN; sends data only to intended recipient using MAC addresses
Router	Connects different networks (e.g., LAN to Internet); routes data using IP addresses
WAP (Wireless Access Point)	Allows wireless devices to connect to a wired network
Transmission media	Physical cables (Ethernet, fibre optic) or wireless signals that carry data

Factors affecting network performance

- **Bandwidth:** Maximum data transfer rate — higher bandwidth = more data can be transferred per second
- **Number of users:** More users = more contention for bandwidth = slower speeds for each user
- **Transmission media:** Fibre optic faster than copper cable; wired faster than wireless
- **Interference:** Wireless signals affected by walls, other devices, distance from access point
- **Network traffic:** Heavy usage times cause congestion and slower speeds
- **Hardware quality:** Older/cheaper equipment may bottleneck the network
- **Error rate:** Damaged cables or interference cause retransmissions, reducing effective speed

Exam tip: When asked about network performance, always consider MULTIPLE factors. A single-factor answer like "bandwidth" won't get full marks. Discuss

> bandwidth, users, interference, and hardware together.

1. Network types and models

A **computer network** is two or more devices connected together to share resources (files, printers, internet connection) and communicate. Networks can be classified by their geographic size (LAN vs WAN) and by how resources are organised (client-server vs peer-to-peer).

1.1 Local Area Network (LAN)

A **LAN** covers a small geographic area, typically a single building, office, school, or home. All the hardware is usually owned and maintained by a single organisation.

Characteristics of a LAN:

- Covers a limited area (one site or building)
- High-speed connections (typically 1 Gbps or faster)
- Hardware owned by the organisation
- Low latency (minimal delay)
- Relatively inexpensive to set up and maintain

Examples: A school's computer network connecting all classrooms, a home WiFi network, an office network connecting employee computers.

Typical LAN hardware: Computers, switches, a router (to connect to the internet), wireless access points, network cables.

1.2 Wide Area Network (WAN)

A **WAN** covers a large geographic area, connecting LANs across cities, countries, or even worldwide. The infrastructure is often provided by third parties like Internet Service Providers (ISPs) or telecommunications companies.

Characteristics of a WAN:

- Covers a wide geographic area (multiple sites, cities, countries)
- Variable speeds depending on connection type
- Often uses leased lines or public infrastructure
- Higher latency than LANs
- More expensive to set up and maintain

Examples: The Internet (the largest WAN, connecting billions of devices worldwide), a bank's network connecting branches across the country, a multinational company's network connecting offices in different countries.

WAN connections: Leased lines (dedicated connections rented from telecoms providers), fibre optic connections, satellite links, or VPNs (Virtual Private Networks) over the public internet to create secure connections between sites.

The Internet as a WAN: The Internet is a global WAN made up of millions of interconnected networks. Your home LAN connects to your ISP's network, which connects to larger networks, forming a hierarchy that spans the globe. When you access a website, data may travel through dozens of networks across multiple countries.

1.3 Client-server model

In a **client-server** network, one or more powerful computers (servers) provide services to other computers (clients). The server manages resources, security, and data centrally.

How it works:

- Clients send requests to the server (e.g., "give me this file")
- Server processes requests and sends responses

- All shared files stored centrally on the server
- User accounts and permissions managed on the server

Types of servers:

- **File server:** Stores and manages shared files
- **Print server:** Manages network printers
- **Web server:** Hosts websites
- **Email server:** Handles sending/receiving emails
- **Database server:** Stores and manages databases

Advantages: Centralised security and backups, easier to manage user accounts, better performance for large networks, files accessible from any client.

Disadvantages: Expensive (server hardware and software), requires specialist IT staff, single point of failure (if server fails, network services unavailable).

1.4 Peer-to-peer model

In a **peer-to-peer (P2P)** network, all computers are equal — there is no central server. Each computer can share its own resources directly with others.

How it works:

- Each computer stores its own files locally
- Computers can access shared resources on other computers
- No central management or control
- Each user responsible for their own security and backups

Advantages: No expensive server required, easy to set up, no reliance on central server (no single point of failure), each computer operates independently.

Disadvantages: Security harder to manage, backups responsibility of each user, performance degrades as network grows, harder to locate files across multiple computers.

> **Exam tip:** The 2024 exam asked candidates to justify which model (client-server or P2P) would be suitable for different scenarios. Always match your answer to the CONTEXT — a school needs client-server for centralised control; a small home network suits P2P.

Worked Example 1 — Describing the difference between LAN and WAN

Question: Describe two differences between a LAN and a WAN. [4 marks]

Model answer:

1. A LAN covers a small geographic area such as a single building or site **(1)**, whereas a WAN covers a large geographic area such as multiple cities or countries **(1)**.
2. A LAN is usually owned and maintained by a single organisation **(1)**, whereas a WAN often uses third-party infrastructure such as ISPs or telecommunications companies **(1)**.

> **Exam tip:** For comparison questions, make sure each point addresses BOTH LAN and WAN. Use comparison words like "whereas" to make the contrast

clear.

Worked Example 2 — Comparing client-server and peer-to-peer

Question: Compare client-server and peer-to-peer networks in terms of security and cost. [4 marks]

Model answer:
Security: Client-server has centralised security managed on the server, making it easier to enforce consistent policies **(1)**. Peer-to-peer has no central security — each user is responsible for their own device's security, making it harder to manage **(1)**.
Cost: Client-server is more expensive because it requires dedicated server hardware and potentially specialist IT staff **(1)**. Peer-to-peer is cheaper because no dedicated server is needed — existing computers share resources **(1)**.

Exam tip: When asked to compare on specific criteria (here: security AND cost), structure your answer clearly around those criteria.

Worked Example 3 — Recommending a network model

Question: A small office has 5 employees who need to share files and a printer. Explain why a peer-to-peer network would be suitable. [3 marks]

Model answer:
A peer-to-peer network is suitable because there are only 5 employees, so the network is small enough that central management isn't essential **(1)**.
It would be cheaper than client-server as no dedicated server hardware or software is required **(1)**.
Each employee can easily share files and the printer directly with others without needing specialist IT support **(1)**.

Exam tip: Always relate your answer to the scenario. Here, the key factors are: small number of users, cost considerations, and simple requirements (just file and printer sharing).

Worked Example 4 — Explaining why client-server is needed

Question: A school has 500 students and 50 staff. Explain why a client-server network would be more appropriate than peer-to-peer. [4 marks]

Model answer:
A client-server network allows centralised management of 550 user accounts, which would be impractical on a peer-to-peer network **(1)**.
Security can be managed centrally — the server can enforce password policies, access permissions, and content filtering for students **(1)**.
Backups can be performed centrally on the server, ensuring student work is protected **(1)**.
Shared resources like files and applications are stored on the server and accessible from any computer in the school **(1)**.

Exam tip: For large organisations like schools, the key arguments are: centralised management, security, backups, and scalability.

2. Network topologies and hardware

A **network topology** describes how devices are arranged and connected in a network. The two topologies you need to know are **star** and **mesh**. You also need to understand the purpose of key hardware components.

2.1 Star topology

In a **star topology**, all devices connect to a central switch or hub. Devices don't connect directly to each other — all communication goes through the central device.

How it works:

- Each device has its own cable to the central switch
- When device A sends data to device B, it goes via the switch
- The switch reads the destination MAC address and forwards data only to the correct device

Advantages of star topology:

- If one cable fails, only that device is affected — others continue working
- Easy to add new devices (just connect to the switch)
- Easy to identify faults (problems isolated to individual connections)
- Better performance than bus topology — no collisions if using a switch
- Centralised management through the switch

Disadvantages of star topology:

- If the central switch fails, the entire network goes down (single point of failure)
- Requires more cable than some other topologies
- Cost of the central switch/hub
- Performance depends on the central device's capacity

Common use: Most modern LANs use star topology because of its reliability and ease of management.

Drawing a star topology: Draw the switch in the centre as a box. Draw each device (computer, printer, etc.) around it. Draw a single line from each device to the central switch. Label each component.

2.2 Mesh topology

In a **mesh topology**, devices are connected to multiple other devices. In a **full mesh**, every device connects to every other device. In a **partial mesh**, devices connect to several (but not all) other devices.

How it works:

- Multiple paths exist between devices
- If one connection fails, data can take an alternative route
- Devices can communicate directly without going through a central point

Advantages of mesh topology:

- Very reliable — no single point of failure
- If one connection fails, alternative routes exist
- Can handle high traffic as multiple paths available
- Data can take the fastest available route

Disadvantages of mesh topology:

- Expensive — requires many connections (full mesh needs n×(n-1)/2 connections for n devices)
- Complex to set up and manage
- Requires more cabling than star topology
- Each device needs multiple network interfaces

Common use: Internet backbone, military networks, wireless mesh networks (home WiFi extenders), situations where reliability is critical.

> **Drawing a mesh topology:** Draw devices in a rough circle or grid. For full mesh, draw lines connecting every device to every other device. For partial mesh, draw connections between most (but not all) pairs.

2.3 Network Interface Card (NIC)

A **NIC** (Network Interface Card or Network Interface Controller) is hardware that allows a device to connect to a network. Every networked device needs a NIC.

Key functions:

- Converts data from the computer into signals for transmission
- Receives signals from the network and converts them to data
- Has a unique **MAC address** (Media Access Control address) — a hardware identifier
- Can be wired (Ethernet port) or wireless (WiFi adapter)

Modern devices (laptops, smartphones) have NICs built into the motherboard. Older computers may have NICs as separate expansion cards.

2.4 Switch

A **switch** connects multiple devices within a LAN. It receives data and forwards it only to the intended recipient.

Key functions:

- Learns which devices are connected to which ports (builds a MAC address table)
- Receives incoming data frames
- Reads the destination MAC address
- Forwards the data only to the correct port — not to all devices

Switch vs Hub: An older device called a hub broadcasts data to ALL connected devices. This wastes bandwidth and causes collisions. Switches are more intelligent — they send

data only to the intended recipient, improving efficiency and security.

2.5 Router

A **router** connects different networks together, typically connecting a LAN to the Internet (a WAN). Routers direct traffic between networks.

Key functions:

- Connects different networks (e.g., home LAN to ISP's network)
- Routes data based on **IP addresses** (not MAC addresses like switches)
- Determines the best path for data to reach its destination
- Often includes firewall functionality for security
- Home routers often combine router, switch, WAP, and modem in one device

How routing works: When you request a website, the router sends your request out to the internet, receives the response, and forwards it back to your device based on your local IP address.

2.6 Wireless Access Point (WAP)

A **WAP** allows wireless devices to connect to a wired network. It acts as a bridge between WiFi devices and the wired LAN.

Key functions:

- Broadcasts a wireless signal (WiFi network) with an SSID (network name)
- Receives wireless connections from devices within range
- Connects to the wired network via Ethernet cable
- Handles encryption (WPA2/WPA3) for security
- Can support multiple devices simultaneously

Note: Home "routers" typically include a built-in WAP, switch, and router in one device. In larger networks like schools or offices, separate WAPs are installed in different locations to provide coverage throughout the building.

2.7 Transmission media

Transmission media are the physical pathways that carry data between devices. The choice of media affects speed, distance, cost, and reliability.

Copper cable (Ethernet):

- Uses electrical signals through twisted pair cables
- Categories: Cat5e (1 Gbps), Cat6 (10 Gbps), Cat7 (10+ Gbps)
- Maximum practical distance: ~100 metres
- Affordable and easy to install
- Susceptible to electrical interference

Fibre optic cable:

- Uses light pulses through glass or plastic fibres
- Very high speeds (100+ Gbps possible)
- Can transmit over many kilometres
- Immune to electrical interference
- More expensive and harder to install than copper
- Used for internet backbone and high-speed connections

Wireless (radio waves):

- No physical cable required
- Affected by distance, obstacles, and interference
- Convenient for mobile devices
- Security requires encryption

Worked Example 5 — Describing star topology

Question: Describe the star network topology and give one advantage and one disadvantage. [4 marks]

Model answer:
Description: In a star topology, all devices are connected to a central switch or hub **(1)**. Devices do not connect directly to each other — all communication passes through the central device **(1)**.
Advantage: If one cable or device fails, only that device is affected and the rest of the network continues to function **(1)**.
Disadvantage: If the central switch fails, the entire network goes down as all communication depends on it (single point of failure) **(1)**.

Exam tip: When describing a topology, explain HOW devices are connected and HOW data flows. Don't just say "devices connect to a switch" — explain that ALL communication goes through the central device.

Worked Example 6 — Explaining the function of a router

Question: Explain the function of a router in a network. [3 marks]

Model answer:
A router connects different networks together, such as a home LAN to the internet **(1)**.
It routes/directs data packets between networks based on IP addresses **(1)**.
It determines the best path for data to reach its destination network **(1)**.

Exam tip: Key points for router: connects DIFFERENT networks, uses IP addresses (not MAC), routes/directs data. Don't confuse with a switch (which connects devices within the SAME network using MAC addresses).

Worked Example 7 — Distinguishing switch from router

Question: Describe the difference between a switch and a router. [4 marks]

Model answer:
A switch connects devices within the same LAN **(1)**, whereas a router connects different networks together (e.g., LAN to internet) **(1)**.
A switch uses MAC addresses to forward data to the correct device within the network **(1)**, whereas a router uses IP addresses to route data between different networks **(1)**.

Exam tip: This is a very common question. Remember: switch = same network, MAC addresses. Router = different networks, IP addresses.

Worked Example 8 — Explaining the purpose of a NIC

Question: Explain the purpose of a Network Interface Card (NIC). [3 marks]

Model answer:
A NIC allows a device to connect to a network **(1)**.
It converts data from the computer into signals that can be transmitted over the network, and vice versa **(1)**.
It has a unique MAC address that identifies the device on the network **(1)**.

Exam tip: Key NIC points: enables connection, converts data to/from signals, has MAC address. Mention that NICs can be wired or wireless if relevant.

3. Wired and wireless connections

Networks can use wired connections (physical cables) or wireless connections (radio waves). Each has advantages and disadvantages, and different technologies are suited to different purposes.

3.1 Wired connections

Wired connections use physical cables to transmit data. The most common types are Ethernet cables (copper) and fibre optic cables.

Ethernet (copper cable):

- Uses twisted pair copper cables (Cat5e, Cat6, Cat7)
- Speeds typically 100 Mbps to 10 Gbps
- Maximum distance about 100 metres
- Affordable and widely used

Fibre optic:

- Uses light pulses through glass/plastic fibres
- Very high speeds (up to 100+ Gbps)
- Can transmit over very long distances
- Immune to electrical interference
- More expensive than copper

Advantages of wired connections:

- Faster and more reliable than wireless
- More secure — harder to intercept (physical access required)
- No interference from other devices
- Consistent speeds regardless of distance (within cable limits)

Disadvantages of wired connections:

- Less convenient — devices must be near network points
- Installation can be expensive and disruptive

- Cables can be trip hazards or unsightly
- Devices are fixed in location
- Not practical for mobile devices

3.2 Wireless connections

Wireless connections use radio waves to transmit data. No physical cables are needed between the device and the access point.

WiFi:

- Based on IEEE 802.11 standards (a/b/g/n/ac/ax)
- Speeds up to several Gbps (WiFi 6/6E)
- Range typically 30-50 metres indoors
- Uses 2.4 GHz and/or 5 GHz frequency bands
- Most common wireless technology for networks

Bluetooth:

- Short-range wireless (typically up to 10 metres)
- Lower power consumption than WiFi
- Used for connecting peripherals (headphones, keyboards, mice)
- Lower speeds than WiFi

Advantages of wireless connections:

- Convenient — connect from anywhere within range
- No cables required — easy to set up
- Supports mobile devices
- Easy to add new devices
- Flexibility in device location

Disadvantages of wireless connections:

- Generally slower than wired
- Less reliable — affected by interference and obstacles
- Security risks — signals can be intercepted (must use encryption)
- Limited range
- Speed decreases with distance from access point
- More users = slower speeds (shared bandwidth)

3.3 Wireless security

Because wireless signals travel through the air, they can potentially be intercepted. **Encryption** is essential to protect wireless networks.

Encryption standards:

- **WEP** (Wired Equivalent Privacy): Old and insecure — easily cracked. Should not be used.
- **WPA** (WiFi Protected Access): Better than WEP but has known vulnerabilities.
- **WPA2**: Current standard — uses strong AES encryption. Recommended.
- **WPA3**: Latest standard — even stronger security. Becoming more common.

Other security measures:

- Strong passwords — long, complex, and unique
- Hide SSID — don't broadcast network name (minor security benefit)
- MAC address filtering — only allow specific devices (can be bypassed)
- Firewall — filter incoming and outgoing traffic
- Regular firmware updates on router/access point

> **Exam tip:** When asked about wireless security, always mention encryption (WPA2 or WPA3). Explain that encryption scrambles data so intercepted signals cannot be read without the key.

3.4 Choosing wired vs wireless

The choice depends on the requirements of the situation:

Choose wired when:

- Speed is critical (gaming, video editing, servers)
- Reliability is essential (business-critical systems)
- Security is paramount (handling sensitive data)
- Devices don't need to move

Choose wireless when:

- Mobility is needed (laptops, smartphones, tablets)
- Running cables is impractical or expensive
- Flexibility in device location is important
- Casual use where maximum speed isn't critical

Many networks use both: Wired connections for desktop computers and servers, wireless for laptops and mobile devices.

3.5 Factors affecting wireless performance

- **Distance:** Signal weakens with distance from access point
- **Obstacles:** Walls, floors, and furniture absorb or reflect signals
- **Interference:** Other WiFi networks, microwaves, Bluetooth devices
- **Number of users:** More devices = shared bandwidth = slower speeds
- **Frequency band:** 5 GHz is faster but shorter range; 2.4 GHz has better range but more interference
- **WiFi standard:** Newer standards (WiFi 6) support higher speeds

Worked Example 9 — Comparing wired and wireless

Question: Compare wired and wireless network connections. Give two advantages of each. [4 marks]

Model answer:

Wired advantages:

1. Faster data transfer speeds than wireless **(1)**.
2. More secure as physical access to the cable is required to intercept data **(1)**.

Wireless advantages:

1. More convenient as devices can connect from anywhere within range without cables **(1)**.
2. Easier to add new devices — no need to install new cables or network points **(1)**.

Exam tip: Keep advantages distinct. "Faster" and "more reliable" can be seen as related. "Faster" and "more secure" are clearly different points.

Worked Example 10 — Explaining WiFi security

Question: Explain how encryption helps secure a wireless network. [3 marks]

Model answer:
Encryption scrambles/encodes the data being transmitted **(1)**.
If the signal is intercepted, the data cannot be read without the encryption key/password **(1)**.
Only devices with the correct key can decrypt and access the data **(1)**.

Exam tip: Explain the PROCESS (data is scrambled) and the EFFECT (intercepted data is unreadable). Mention that a key/password is needed to decrypt.

Worked Example 11 — Recommending a connection type

Question: A graphic designer works with large video files. Explain why a wired connection would be more suitable than wireless for this work. [3 marks]

Model answer:
Wired connections provide faster data transfer speeds, allowing large video files to be transferred more quickly **(1)**.
Wired connections are more reliable with consistent speeds, which is important when editing or rendering video **(1)**.
Wired connections have lower latency, which improves responsiveness when working with large files on a network **(1)**.

Exam tip: Relate each point to the scenario. "Faster" alone isn't enough — explain WHY faster matters for video work.

Worked Example 12 — Factors affecting network performance

Question: Describe three factors that can affect the performance of a wireless network. [6 marks]

Model answer:

1. Distance from access point: The further a device is from the wireless access point, the weaker the signal **(1)**. This results in slower speeds and potential disconnections **(1)**.
2. Physical obstacles: Walls, floors, and other obstacles can absorb or reflect wireless signals **(1)**. This reduces signal strength and can create "dead zones" with no coverage **(1)**.
3. Number of users: More devices connected to the network means the available bandwidth is shared between them **(1)**. This reduces the speed available to each individual device **(1)**.

Exam tip: For "describe" questions worth 2 marks per point, give the factor AND explain its effect. Just listing factors won't get full marks.

Worked examples summary

The 12 worked examples above cover the main question types you will encounter on networks. Here is a summary of the key question patterns:

Example	Question type	Key skill
1	LAN vs WAN (4 marks)	Geographic area + ownership comparison
2	Client-server vs P2P (4 marks)	Security + cost comparison
3	Recommend P2P (3 marks)	Small office justification
4	Recommend client-server (4 marks)	Large organisation justification
5	Star topology (4 marks)	Description + advantage/disadvantage
6	Router function (3 marks)	Connects networks, uses IP addresses
7	Switch vs router (4 marks)	Network scope + address type comparison
8	NIC purpose (3 marks)	Connection + conversion + MAC address
9	Wired vs wireless (4 marks)	Two advantages of each
10	Encryption explanation (3 marks)	Scrambles data, requires key
11	Connection recommendation (3 marks)	Relate to specific scenario
12	Wireless performance factors (6 marks)	Factor + effect for each

Quick reference

Key terms — exam-ready definitions

Term	Definition (mark scheme style)
LAN	Local Area Network; covers a small geographic area (single building/site); usually owned by one organisation
WAN	Wide Area Network; covers a large geographic area (cities/countries); often uses third-party infrastructure
Client-server	Network model where central server(s) provide services to client computers; centralised management
Peer-to-peer	Network model where all computers are equal; share resources directly; no central server
Star topology	Network layout where all devices connect to a central switch; single point of failure at switch
Mesh topology	Network layout where devices connect to multiple other devices; no single point of failure
NIC	Network Interface Card; allows device to connect to network; has unique MAC address
Switch	Connects devices within a LAN; forwards data to correct device using MAC addresses
Router	Connects different networks; routes data using IP addresses; typically connects LAN to Internet
WAP	Wireless Access Point; allows wireless devices to connect to wired network; broadcasts WiFi signal
MAC address	Media Access Control address; unique hardware identifier assigned to each NIC
IP address	Internet Protocol address; logical address that identifies a device on a network; can change
Bandwidth	Maximum data transfer rate of a network connection; measured in bits per second
Encryption	Scrambling data so it cannot be read without the correct key; protects wireless transmissions
The Internet	A worldwide collection of interconnected networks; the largest WAN
WiFi	Wireless networking technology using radio waves; based on IEEE 802.11 standards

Term	Definition (mark scheme style)
Bluetooth	Short-range wireless technology for connecting peripherals; lower power than WiFi
Ethernet	Wired networking standard using copper cables; common for LANs
Fibre optic	Transmission media using light pulses through glass/plastic; very high speed, long distance
SSID	Service Set Identifier; the name of a wireless network broadcast by a WAP

Hardware comparison

Device	Connects	Uses	Purpose
NIC	Device to network	MAC address	Enable network connection
Switch	Devices in same LAN	MAC addresses	Forward data within LAN
Router	Different networks	IP addresses	Route data between networks
WAP	Wireless to wired	WiFi signals	Enable wireless connections

Topology comparison

Feature	Star	Mesh
Structure	All devices → central switch	Devices → multiple other devices
Single point of failure	Yes (central switch)	No (multiple paths)
Cost	Lower	Higher (more connections)
Complexity	Simple	Complex
Common use	Most LANs	Internet backbone, critical systems

Typical pitfalls

Pitfall 1 — Confusing switch and router.
Wrong: "A switch connects the LAN to the internet."
Right: A **switch** connects devices within the same LAN. A **router** connects different networks (like LAN to internet).

Pitfall 2 — Confusing MAC and IP addresses.
Wrong: "A router uses MAC addresses to send data."
Right: Routers use **IP addresses**. Switches use **MAC addresses**. MAC = hardware, IP = network location.

Pitfall 3 — Incomplete topology descriptions.
Wrong: "In a star, devices connect to a switch."
Right: "All devices connect to a **central** switch, and all communication passes through this central point."

Pitfall 4 — Single-sided comparisons.
Wrong: "A LAN covers a small area."
Right: "A LAN covers a small area **whereas** a WAN covers a large geographic area."

Pitfall 5 — Forgetting context in recommendations.
Wrong: "Wired is better because it's faster."
Right: "Wired is better for this scenario because the video editor needs fast speeds **to transfer large files quickly**."

Pitfall 6 — Thinking wireless is always worse.
Wrong: "Wireless is bad because it's slower and less secure."
Right: Wireless has **advantages** (convenience, mobility, easy setup) that make it appropriate for many situations.

Pitfall 7 — Vague security answers.
Wrong: "Encryption makes the network secure."
Right: "Encryption **scrambles data** so that if signals are intercepted, they **cannot be read without the key**."

Pitfall 8 — Confusing server types with network models.
Wrong: "A file server is a type of network."
Right: A file server is a **type of server** in a client-server **network model**.

Pitfall 9 — Not explaining WHY factors affect performance.
Wrong: "Distance affects performance."
Right: "Greater distance means weaker signal strength, **resulting in** slower speeds and potential disconnections."

Pitfall 10 — Forgetting mesh topology advantages.
Wrong: "Mesh is expensive so it's never used."

Right: Mesh is used where **reliability is critical** (internet backbone, military) because there's no single point of failure.

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